

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 08



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A stylized illustration of a laptop with a blue frame and an orange base. The screen is white and displays the text 'TOPICS TO BE COVERED' in a green, hand-drawn font.

TOPICS TO BE COVERED

01 Basics of Generating function

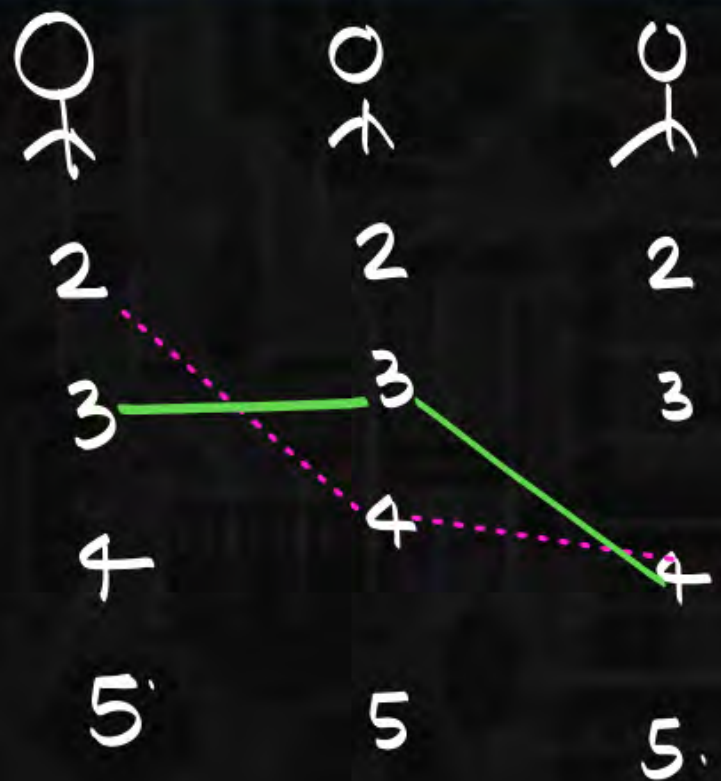
02 Problems on Generating function

COMBINATORICS

How many ways we can distribute
 10 similar coins among 3 children
 such that each child gets at least 2 coins & at most 5 coins?

COMBINATORICS

{ at least 2
at most 5.



$$\underline{2} + \underline{4} + \underline{4} = 10 \quad (\text{at least 2} \\ \text{at most 5})$$

$$\underline{3} + \underline{4} + \underline{4} = 10.$$

$$n^2 n^4 n^4 = n^{10}.$$

(12) no.

$$\begin{cases} 244 \\ 424 \\ 442 \end{cases}$$

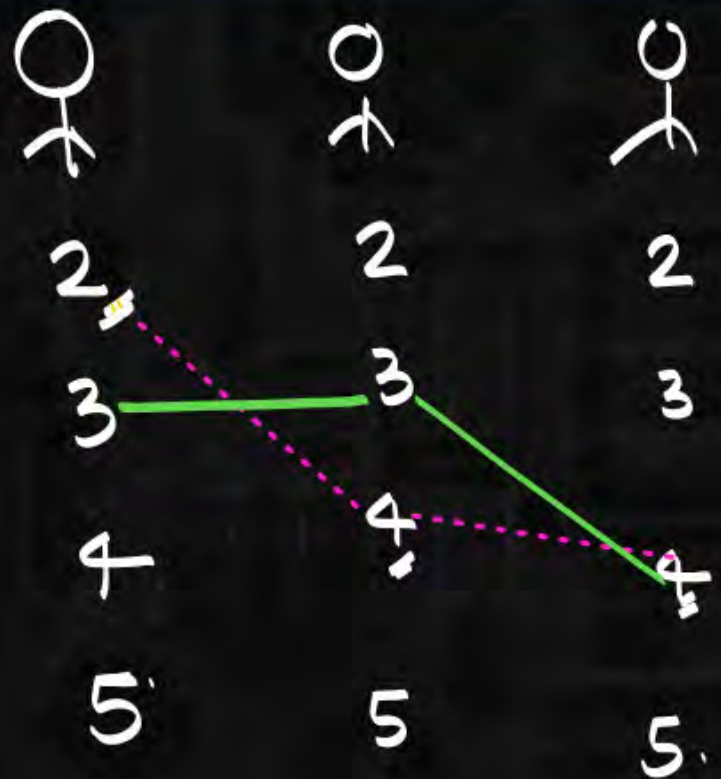
$$\begin{cases} 523 \\ 253 \\ 532 \\ 235 \\ 352 \\ 325 \end{cases}$$

$$\begin{cases} 463 \\ 643 \\ 334 \end{cases}$$

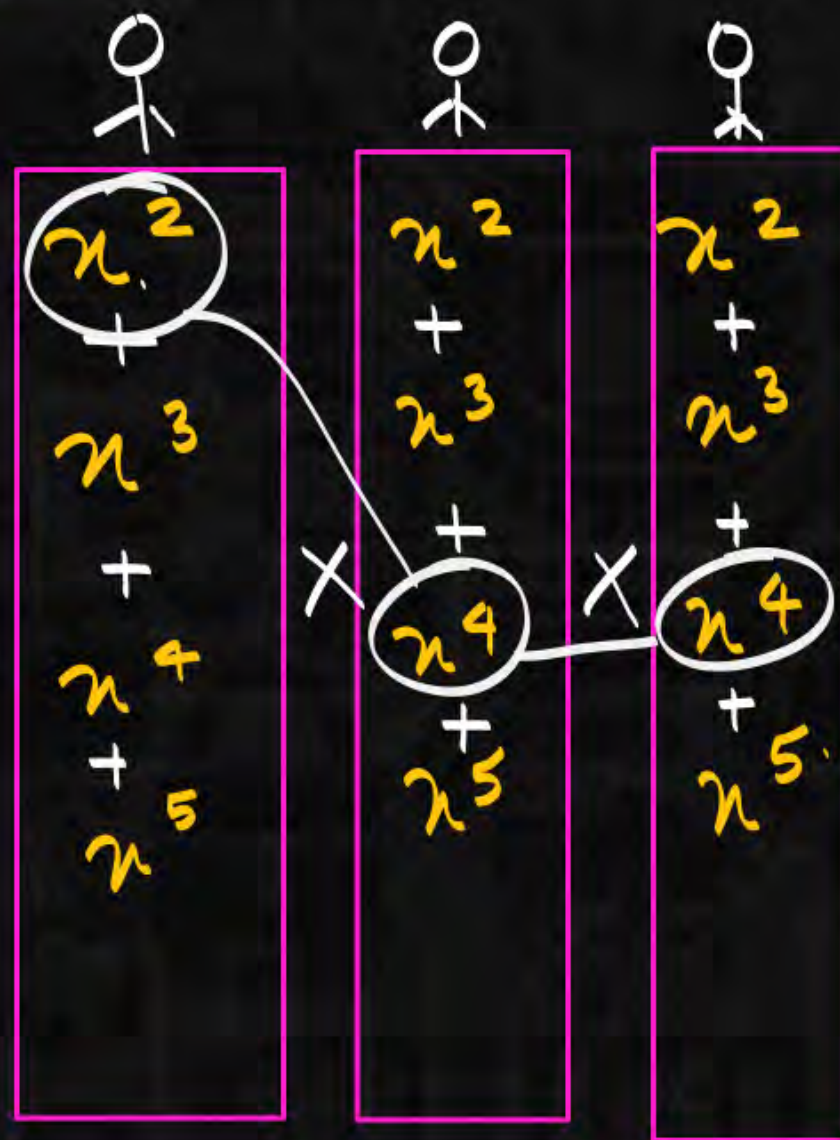
COMBINATORICS



$\left\{ \begin{array}{l} \text{at least 2} \\ \text{at most 5} \end{array} \right.$



Giving 2 coins = x^2
 Giving 3 coins = x^3
 ...



$$\rightarrow x^2 x^4 x^4 = x^{10}$$

$$x^{e_1} x^{e_2} x^{e_3} = x^{10}$$

$$\left\{ \begin{array}{l} e_1 + e_2 + e_3 = 10 \\ \text{at least 2} \\ \text{at most 5} \end{array} \right.$$

COMBINATORICS



$$(x^2 + x^3 + x^4 + x^5) \times (x^2 + x^3 + x^4 + x^5) \times (x^2 + x^3 + x^4 + x^5) \xrightarrow{x^{10}}$$

$$(x^2 + x^3 + x^4 + x^5)^3$$

finding the coefficient of x^{10}

$$(x^2 + x^3 + x^4 + x^5)^3$$

$$(x^2)^3 (1 + x + x^2 + x^3)^3$$

$$x^6 (1 + x + x^2 + x^3)^3$$

COMBINATORICS



$$x^6 (1+x+x^2+x^3)^3$$

$$x^6 \left(\frac{1-x^4}{1-x} \right)^3$$

$$x^6 (1-x^4)^3 (1-x)^{-3}$$

$$x^6 (1-3x^4+3x^8-x^{12}) (1-x)^{-3}$$

$$1+x+x^2+x^3 = \frac{x^4-1}{x-1} = \frac{1-x^4}{1-x}$$

$$x^6 (1-3x^4+3x^8-x^{12}) (1-x)^{-3}$$

$$(x^6-3x^{10}+3x^{14}-x^{18}) (1-x)^{-3}$$

COMBINATORICS



$$(x^6 - 3x^{10} + 3x^{14} - x^{18})(1-x)^{-3}$$

$$1+x+x^2+x^3 = \frac{x^4-1}{x-1} = \frac{1-x^4}{1-x}$$

$$x^6(1-3x^4+3x^8-x^{12})(1-x)^{-3}$$

$$(x^6 \times 15x^4 - 3x^{10})$$

$$(15-3)x^{10}$$

$$12x^{10}$$

coefficient of x^4

$$-3c_4(-x)^4$$

$$(-1)^4 \frac{3+4-1}{c_4} x^4$$

$$6c_4(-x)^4$$

$$15x^4$$

$$(x^6 - 3x^{10} + 3x^{14} - x^{18})(1-x)^{-3}$$

COMBINATORICS

coefficient of x^{10}

$$(x^6 - 3x^{10} + 3x^{14} - x^{18})(1-x)^{-3}$$

$$(x^6 - 3x^{10} + 3x^{14} - x^{18}) \left[-3C_0(-x)^0 + -3C_1(-x)^1 + -3C_2(-x)^2 + -3C_3(-x)^3 + -3C_4(-x)^4 \right]$$

$$x^6 \times -3C_4(-x)^4 - 3x^{10} \cdot 1$$

$$15x^{10} - 3x^{10}$$

$$(15 - 3)x^{10} = 12x^{10}$$

$$-3C_4 = (-1)^4 3+4-1C_4$$

$$= 1 \cdot 6C_4 = 15$$

$$(x^2 + x^3 + x^4 + x^5)^3$$

take x^2 .

$$(x^2)^3 [1 + x + x^2 + x^3]^3$$

$$x^6 \left(\frac{1 - x^4}{1 - x} \right)^3$$

$$x^6 (1 - x^4)^3 (1 - x)^{-3}$$

$$x^6 (1 - 3x^4 + 3x^8 - x^{12}) (1 - x)^{-3}$$

$$(x^6 - 3x^{10} + 3x^{14} - x^{18}) (1 - x)^{-3}$$

$$(x^6 x - 3 {}_4C_4 (-x)^4 - 3x^{10})$$

$$x^6 x (-1)^4 (3 + 4 - 1) {}_4C_4 (x^4) - 3x^{10}$$

$$\therefore \downarrow 15x^{10} - 3x^{10} = 12x^{10} \downarrow$$

COMBINATORICS



$$(a+b)^n = \sum_{i=0}^n {}^nC_i \times a^{n-i} \times b^i$$

$$-{}^nC_k = (-1)^k {}^{n+k-1}C_k$$

$$-{}^3C_1 = (-1)^1 {}^{3+1-1}C_1 =$$

$$n=3 \quad k=1$$

$$\left(\frac{1}{1-x}\right)^3 = (1+(-x))^{-3}$$

$$= -{}^3C_0 (1)^{-3-0} (-x)^0 + \underbrace{-{}^3C_1 (1)^{-3-1} (-x)^1}_{\substack{\downarrow \text{ -ve} \\ \text{true}}} + \underbrace{-{}^3C_2 (1)^{-3-2} (-x)^2}_{\substack{\downarrow \text{ -ve} \\ \text{true}}} + \underbrace{-{}^3C_3 (1)^{-3-3} (-x)^3}_{\substack{\downarrow \text{ -ve} \\ \text{true}}}$$

COMBINATORICS

How many ways we can distribute
12 similar coins among 3 students.
so that each student gets at least 2
at most 6 coins.


 2
 3
 4
 5
 6


 2
 3
 ...
 6


 2
 ...
 6

$$\begin{pmatrix} x^2 \\ x^3 \\ x^4 \\ \vdots \\ x^6 \end{pmatrix}$$

$\times$

$$\begin{pmatrix} x^2 \\ x^3 \\ + \\ \vdots \\ x^6 \end{pmatrix}$$

$\times$

$$\begin{pmatrix} x^2 \\ + \\ x^3 \\ + \\ \vdots \\ x^6 \end{pmatrix}$$

COMBINATORICS



$$(x^2 + x^3 + x^4 + \dots + x^6)^3$$

take x^2

$$(x^2)^3 [1 + x + x^2 + x^3 + x^4]^3$$

$$x^6 \left[\frac{1-x^5}{1-x} \right]^3$$

$$-3c_6 (-x)^6$$

$$(-1)^6 3+6-1 c_6 (-x)^6$$

coefficient of x^{12} .

$$x^6 [1-x^5]^3 [1-x]^{-3}$$

$$x^6 [1-3x^5+3x^{10}-x^{15}] [1-x]^{-3}$$

$$(x^6 - 3x^{11} + 3x^{16} - x^{21}) (1-x)^{-3}$$

$$\left(x^6 \times -3c_6 (-x)^6 - 3x^{11} \times -3c_1 (-x)^1 \right)$$

$$8c_6 (x)^{12} - 3 \times 3 x^{12}$$

$$= 19$$

15 coins
+ children
at least 3
at most 5.

Ans: 1.

$$(x^3 + x^4 + x^5)^4$$

$$(x^3)^4 \cdot [1 + x + x^2]^4$$

$$x^{12} [1 + x + x^2]^4$$

$$x^{12} \left[\frac{1 - x^3}{1 - x} \right]^4$$

$$= x^{18}$$

$$x^{12} [1 - x^3]^4 [1 - x]^{-4}$$

$$x^{12} [1 - 4x^3 + 6x^6 - 4x^9 + x^{12}] [1 - x]^{-4}$$

$$x^{12} [1 - 4x^3 + 6x^6 - 4x^9 + x^{12}] (1-x)^{-4} \quad \text{--- } x^{15}$$

$$x^{12} [1 - x^3]^4 (1-x)^{-4}$$

$$\left[\frac{x^{12}(1) \times -4}{C_3} (-x)^3 - 4 \times x^{15} \right]$$

$$x^{12} [1 - 4x^3 + 6x^6 - 4x^9 + x^{12}] (1-x)^{-4}$$

$$x^{12} \cancel{(-1)^3}^{4+3-1} C_3 \cancel{(-1)^3} x^3$$

$$6C_3 x^{15} - 4x^{15} \quad (20-4)x^{15} = 16x^{15}$$

COMBINATORICS



coefficient of x^{12} .

$$\left(x^3 + x^4 + x^5 + x^6 \cdot \infty \right)^3$$

$$(x^3)^3 \left(\underline{1 + x + x^2 + x^3} \cdot \infty \right)^3$$

$$x^9 \left(\frac{1}{1-x} \right)^3$$

$$x^9 (1-x)^{-3}$$

$$x^9 \left(-3C_3 (-x)^3 \right)$$

$$3+3-1C_3$$

Ans: 10

COMBINATORICS



$$\frac{1}{1-x} = 1 + x + x^2 + x^3 \dots \infty$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 \dots \infty$$

$$\frac{1}{1-ax} = 1 + ax + (ax)^2 + (ax)^3 + (ax)^4 \dots \infty \rightarrow a=2 \quad \frac{1}{1-2x} = 1 + 2x + (2x)^2 + (2x)^3$$

$$\frac{1}{1+ax} = 1 - ax + (ax)^2 - (ax)^3 + (ax)^4 \dots \rightarrow a=2 \quad \frac{1}{1+2x} = 1 - 2x + (2x)^2 - (2x)^3$$

$$1 + x + x^2 = \frac{1-x^3}{1-x}$$

$$\left\{ 1 + x + x^2 \dots x^n = \frac{1-x^{n+1}}{1-x} \right.$$

COMBINATORICS



24 coins

4 students

at least 3

at most 8

Ans: 125.

Coefficient of x^{50}

$$(x^7 + x^8 + x^9 + \dots + \infty)^6$$

coefficient of x^{20}

$$(x^2 + x^3 + x^4 + x^5 + x^6)^5$$

